



CISCO NETWORKING ACADEMY – VIRTUAL INTERNSHIP
PROGRAM

NetSage AI: Network Troubleshooting & Diagnostics

A 9-Week AI-Assisted Networking Internship Report

INTERNSHIP DURATION

May 15 – July 15, 2026 (9 Weeks)

STUDENT CREDENTIALS

Veer Gupta

Enrollment / Roll No.

25SCS1003003599

Program & Degree

B.Tech (CSE)

HOST PROGRAM

Cisco Networking Academy

In partnership with IILM University

Candidate & Program Overview

Candidate Details

Student Name: VEER GUPTA

Roll Number: 25SCS1003003599

Degree: B.Tech (Computer Science & Engineering)

Academic Batch: 2025-2029

University: IILM University, Greater Noida, U.P.

Internship Profile

Program Name: Cisco Networking Academy (VIP)

Specialization Domain: AI Network Troubleshooting

Duration: 9 Weeks (May 15 – July 15, 2026)

Certificate ID: c2681fa6...c9de9 (see Slide 11)

Instructor: Swati Vashisht, Cisco Networking Academy

Organization Profile: Cisco Networking Academy

Cisco Networking Academy is a global IT skills program delivered through IILM University, operating under the vision of *"Bridging Classrooms and Careers in Technology"*.

 **Cisco Collaboration:** Delivered through IILM University in partnership with Cisco Networking Academy to empower engineering talent across India.

 **Industry Alignment:** Partners with Cisco's global academy network to bridge the gap between classroom learning and industry-ready networking and AI skills.

 **Core Focus Domains:** Networking Fundamentals, Data Analytics, Applied Artificial Intelligence, and Cybersecurity.

 Robotic Process Automation and AI Dashboard

Problem Statement & Industry Context

The Industry Challenge

Students and junior network engineers can often run the right networking commands but struggle to connect a symptom to its true underlying cause among many possible faults.

A connectivity issue could stem from a VLAN misconfiguration, gateway mismatch, DHCP failure, routing problem, DNS issue, ACL rule, or NAT misconfiguration.

The Solution Scope

This internship builds NetSage AI, a decision-support system that analyzes troubleshooting evidence and proposes a probable root cause and next diagnostic step.

Combining a deterministic Python rule engine with Gemini-powered NLP diagnosis, while always keeping a human reviewer in the loop before any fix is accepted.

Core Project Objectives



Rule-Based Diagnosis

Build a deterministic Python rule engine that detects known network faults such as VLAN, DHCP, and gateway issues from evidence patterns.



NLP / AI Diagnosis

Use Gemini-powered similarity matching to rank Top-3 candidate diagnoses when no deterministic rule matches the evidence.



Hybrid Decision & Review

Combine rule-based and AI outputs into a High-Confidence or Human-Review verdict, with Accept, Edit, or Reject controls.



Responsible AI Analytics

Track agreement rate, category/severity breakdowns, and a full AI-vs-human corrections audit log on a Streamlit dashboard.

Technologies & Tech Stack



Dashboard & UI

Language: Python (core application logic)

UI Framework: Streamlit dashboard and review panel



Diagnosis Engine

AI Engine: Google Generative AI (Gemini API)

Rule Engine: Deterministic Python rule-based checks



Data & Config

Data Handling: pandas for case datasets & logs

Configuration: python-dotenv for environment secrets



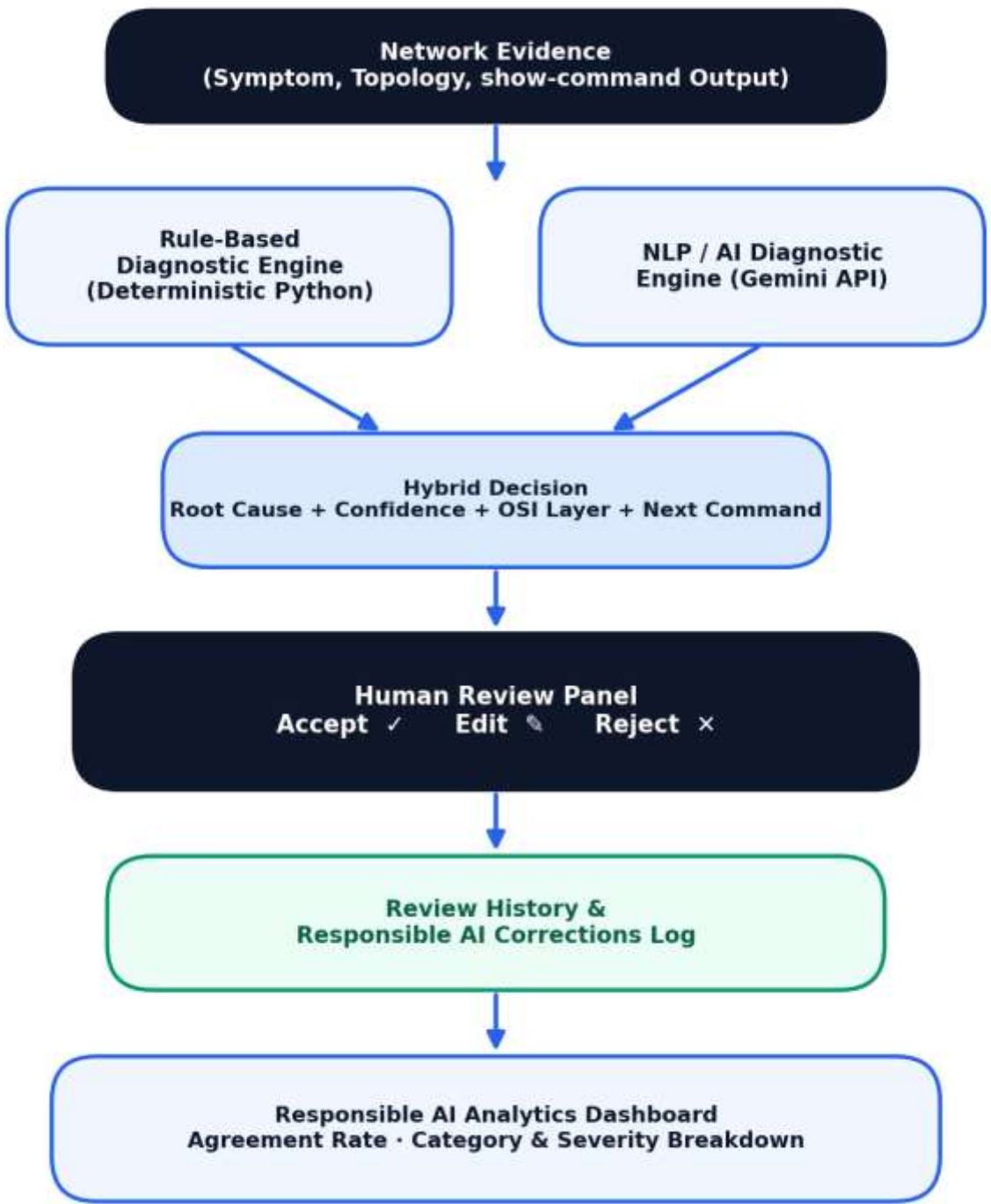
Analytics & Logging

Dashboard: Responsible AI analytics & agreement rate

Audit Log: Review History & Corrections Log

System Architecture & Workflow

-  **Evidence Layer:** Symptom, topology, and Cisco show-command output submitted via the dashboard.
-  **Rule Engine:** Deterministic Python engine matches evidence against known network fault patterns.
-  **AI / NLP Engine:** Gemini-powered similarity matching ranks Top-3 candidate diagnoses.
-  **Hybrid Decision:** Combines both engines into a High-Confidence or Human-Review verdict.
-  **Human Review & Logging:** Reviewer accepts, edits, or rejects; every decision is logged for audit.



Application Walkthrough

Troubleshooting Desk – Case Selection & Evidence

Deploy ⋮

 **NETSAGE AI: Troubleshooting Helper**

Troubleshooting Desk Analytics Dashboard

1. Select a Case

Choose a networking problem:

CASE-001 - PC1 cannot ping PC2 ⌵

Case Details

Symptom: PC1 cannot ping PC2

Topology: PC1 in VLAN 10, PC2 in VLAN 20. Router on a stick configured.

🗨️ Run AI Diagnosis

show_command_output (Show Commands)

```
Router: show ip int brief -> GigabitEthernet0/0.10 up up, GigabitEthernet0/0.20
```


Application Walkthrough

AI Diagnosis & Human-in-the-Loop Review

Deploy

2. AI Diagnosis & Human Review

Root Cause: The physical port mapped to VLAN 40 on the Wireless LAN Controller (WLC) is down, breaking network connectivity and preventing clients on the 'Corp' SSID from reaching the DHCP server.

Confidence: High | OSI Layer: Layer 1

Evidence: WLC: show interface summary → interface vlan40 physical port down

Next Command:

show port summary

Fix Steps:

Inspect physical cabling and port link status on the WLC and connected switch. Re-enable the physical interface using the 'no shutdown' command if administratively down.

⚠ System requires human review before applying fix.

Human Review Panel

Review Decision

✔ Accept Diagnosis

✎ Edit Diagnosis

✖ Reject Diagnosis

Application Walkthrough

Review Submission: Edit, Correct & Log Decisions

Deploy

Human Review Panel

Review Decision

✓ Accept Diagnosis

✎ Edit Diagnosis

✗ Reject Diagnosis

Current Status: Edited

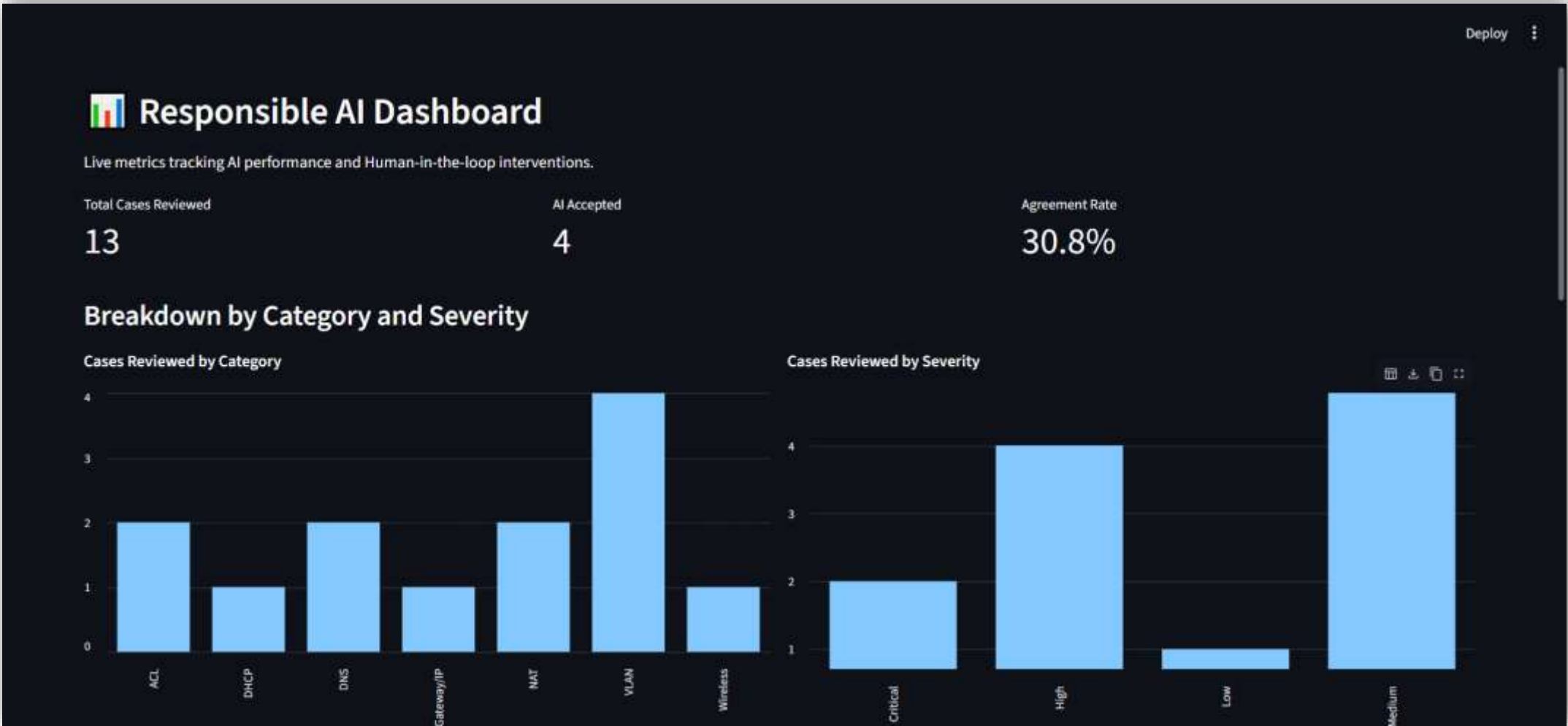
Corrected Root Cause (if applicable)

Reason for Correction/Rejection

Submit Review to Logs

Application Walkthrough

Responsible AI Analytics Dashboard



Application Walkthrough

Decision Breakdown by Category

Deploy ⋮

category	Accepted	Edited	Rejected
ACL	1	1	0
DHCP	0	1	0
DNS	0	2	0
Gateway/IP	0	0	1
NAT	1	1	0
VLAN	1	2	1
Wireless	1	0	0

Review History

	Timestamp	Case_ID	Decision	Reason
0	2026-08-23 21:10:30	CASE-001	Edited	Testing the Responsible AI logging system
1	2026-08-26 01:29:19	CASE-027	Accepted	None
2	2026-08-26 01:31:28	CASE-008	Rejected	None
3	2026-08-26 01:32:27	CASE-004	Rejected	didn't work for me
4	2026-08-26 01:32:57	CASE-028	Accepted	None
5	2026-08-26 01:35:05	CASE-015	Edited	When correcting an outdated A record in a DNS zone file, errors or rejections usually happen because of stric
6	2026-08-26 01:45:36	CASE-002	Edited	show int trunk on both Sw1 and Sw2 shows Mode auto / Status not-trunking. Two auto ports never negotiate

Application Walkthrough

Review History & Responsible AI Corrections Log

Deploy

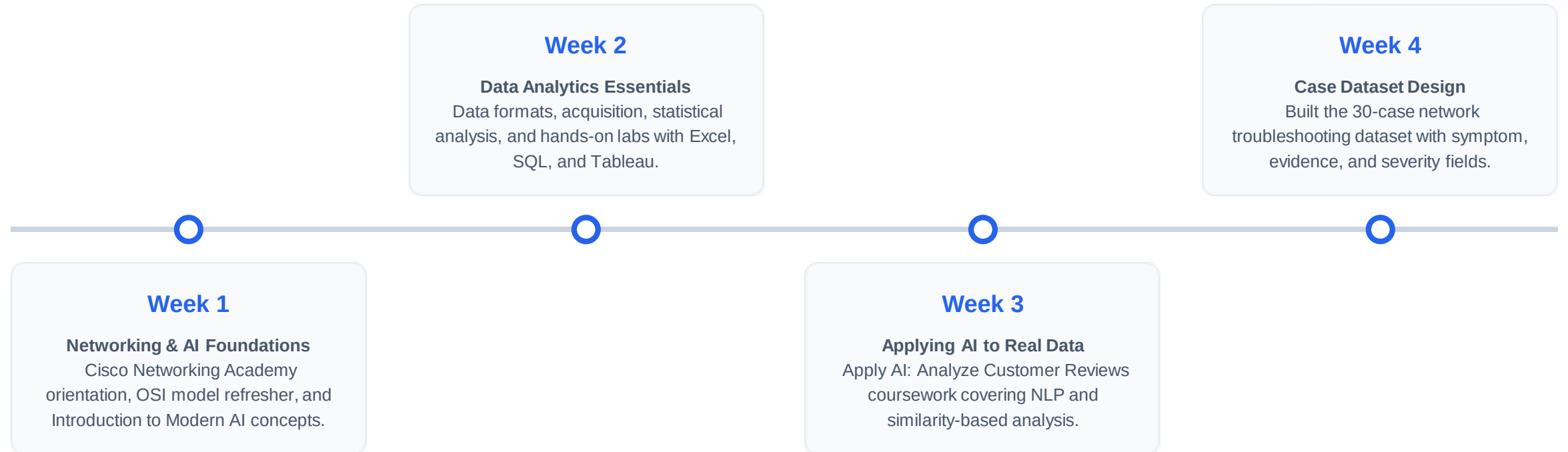
Review History

	Timestamp	Case_ID	Decision	Reason
3	2026-08-26 01:32:27	CASE-004	Rejected	didn't work for me
4	2026-08-26 01:32:57	CASE-028	Accepted	None
5	2026-08-26 01:35:05	CASE-015	Edited	When correcting an outdated A record in a DNS zone file, errors or rejections usually happen because of stric
6	2026-08-26 01:45:36	CASE-002	Edited	show int trunk on both Sw1 and Sw2 shows Mode auto / Status not-trunking. Two auto ports never negotiate
7	2026-08-26 01:46:32	CASE-009	Edited	The PC and DHCP server are on different subnets, and DHCP relies on broadcast. Without ip helper-address c
8	2026-08-26 01:47:15	CASE-013	Edited	ipconfig shows DNS Servers: 127.0.0.1. Since the PC can already ping 8.8.8.8 by IP, connectivity is fine — the i
9	2026-08-26 01:47:53	CASE-021	Edited	show access-lists 10 shows only a permit for 10.1.1.0/24. Standard ACLs end with an implicit deny any, so any
10	2026-08-26 01:48:40	CASE-025	Edited	Gi0/1 correctly has ip nat inside, but the show run output for Gi0/0 shows no NAT command applied at all. N
11	2026-08-26 02:08:25	CASE-003	Accepted	None
12	2026-08-26 02:08:48	CASE-023	Accepted	None

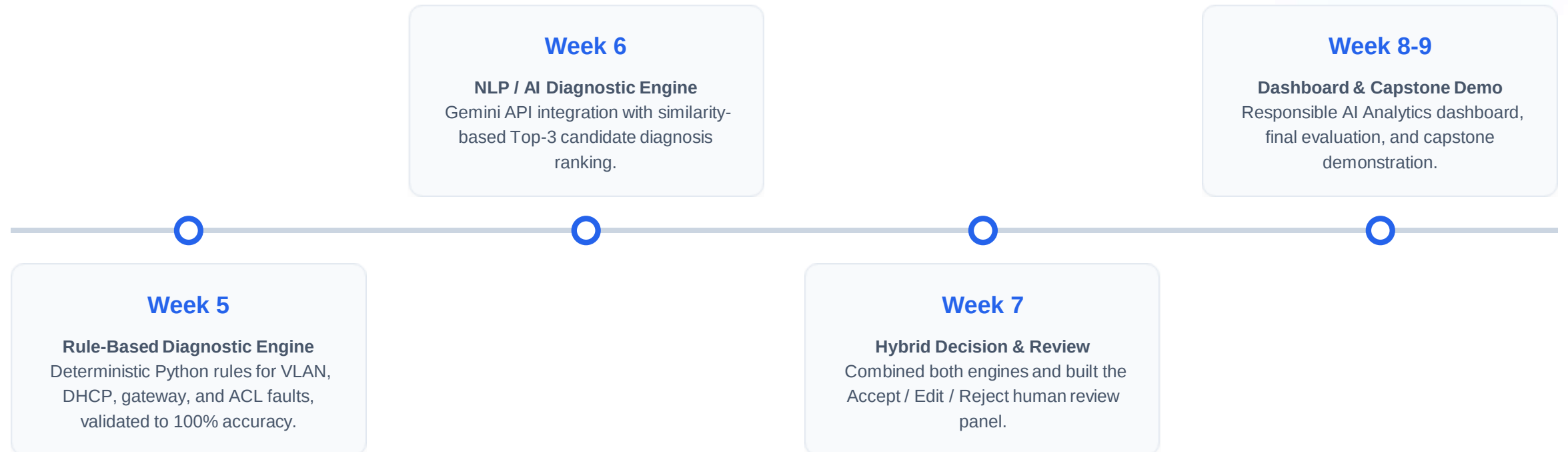
 Responsible AI Corrections Log

	Timestamp	Case_ID	AI_Diagnosis	Human_Correction	Reason
0	2026-08-23 21:10:30	CASE-001	{'root_cause': 'The router subinterface GigabitEthernet0/0.20 serving as the default g	{'root_cause': 'The router subinterface GigabitEthernet0/0.20 serving as the default g	Testing the Respo
1	2026-08-26 01:31:28	CASE-008	Serial0/0/0 interface status is 'up/down', indicating a Layer 2 Data Link layer failure dt	None	None
2	2026-08-26 01:32:27	CASE-004	Native VLAN mismatch on trunk interface Gi0/1 (Sw1 uses VLAN 99 while Sw2 uses VL	None	didn't work for m

Curriculum Timeline (Weeks 1 to 4)



Curriculum Timeline (Weeks 5 to 9)



Key Projects & Demonstrated Skills

- ✓ **Rule-Based Diagnostic Engine:** Built a deterministic Python rule engine validated to 100% accuracy across all 30 troubleshooting cases.
- ✓ **NLP/AI Diagnostic Engine:** Implemented Gemini-powered similarity matching, achieving 83.33% Top-1 and 90.00% Top-3 accuracy on unseen evidence.
- ✓ **Full-Stack Streamlit Dashboard:** Built an end-to-end Troubleshooting Desk and Analytics Dashboard using Streamlit and pandas.
- ✓ **Human-in-the-Loop Review System:** Designed an Accept / Edit / Reject review workflow that logs every reviewer decision and correction reason.
- ✓ **Responsible AI Analytics:** Built category/severity breakdown charts and a Responsible AI Corrections Log to track AI-human agreement over time.

Certification & Official Verification

Verification Metric	Official Record / Credential Details
Internship Title	AI-Assisted Network Troubleshooting Virtual Internship
Issuing Organization	Cisco Networking Academy (delivered via IILM University)
Academic Patronage	IILM University – School of Computer Science & Engineering
Certificate 1 – Intro to Modern AI	d08b1949-9730-46c9-b3f5-81e6244cb417 (Issued: Jun 3, 2026)
Certificate 2 – Apply AI: Reviews	c2681fa6-2d20-4047-901d-ba7afc1c9de9 (Issued: Jun 5, 2026)
Certificate 3 – Data Analytics	Data Analytics Essentials, verified by QR (Issued: Jun 12, 2026)

Certification & Official Verification

Certificate 1 of 3 – Introduction to Modern AI



Certification & Official Verification

Certificate 2 of 3 – Apply AI: Analyze Customer Reviews



Certification & Official Verification

Certificate 3 of 3 – Data Analytics Essentials



Thank You!



Veer Gupta

Roll No: 25SCS1003003599



**School of Computer Science &
Engineering**

IILM University, Greater Noida, U.P.